

WR-IX

Epistemic Horizons: What Can an Observer Inside a World Ever Distinguish?

Accessible protocols, resource certificates
and future-dependent causal reach
Mathematical draft v0.1

A. A. Malachevsky

ORCID: 0009-0008-6009-3196

World Realizability & Epistemic Horizons (WREH)

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Abstract

A difference between models can remain unknown, fail to be observable with present resources, or fail to produce any response under an entire declared future protocol class. These are different statements. We attach explicit time and energy budgets to observer protocols and give exact and bounded-error separation certificates, while retaining the response equivalence inherited from WR-I. A finite adaptive transcript cannot distinguish models whose conditional response laws agree at every common feasible history. The concrete benchmark is an ideal tagged light pulse in a flat expanding cubic torus, compared with its Euclidean cover. Its first return exists exactly when the torus period is strictly smaller than the remaining comoving light-travel span. A detector threshold adds a separate energy condition. For exponential expansion we derive the exact time–energy detection frontier, including its unattained causal endpoint. Smooth positive expansion histories can agree on the entire supplied past and on an arbitrarily long finite future, yet have finite versus infinite remaining conformal span. Thus current background compatibility does not determine permanent future accessibility without additional dynamical assumptions. The ninth paper closes the first technical cycle at the level of a reviewable mathematical draft; it establishes no universal limit on knowledge or new cosmological law.

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1 The final obligation of the first cycle

The WREH manifesto assigns WR-IX the distinction between unknown, currently unobservable, inaccessible in principle and physically indistinguishable [1]. WR-I already owns response equivalence and finite separation on compact domains [2]; WR-II–VI own existence, fibre geometry, historical and experimental completion, and coupled admissibility [3, 4, 5, 6, 7]. WR-VII supplies a calibrated unit-bearing quantum resource benchmark, rather than a universal observer resource bound [8]. WR-VIII supplies a finite current cosmological patch and alternative kinematic lifts [9].

The new obligation is to construct an observer-accessible future response family and its resource frontier, then test whether the frontier is determined by current data. We do not relabel the WR-I quotient

as a new result, identify refinement order with physical time, or import the qubit carrier into cosmology. The mathematical arguments below are elementary and self-contained; no theorem-priority claim or exhaustive novelty certificate is made.

2 Access records and the scope of a horizon

Fix a nonempty current compatibility class C , an observer/laboratory description O , and a class $\mathcal{P}$ of complete finite protocols. Each protocol has a common response map $r_p : C \rightarrow \mathbb{R}$ and certified upper bounds $t_p \geq 0$, $e_p \geq 0$ on elapsed proper time and launch energy. Units are seconds and joules. Scalar responses are normalized in stated response units. For this first abstract interface, feasibility and the certified bounds hold for every $W \in C$.

$$\mathcal{A}_{\tau,E} = \{p \in \mathcal{P} : t_p \leq \tau, e_p \leq E\}, \quad \mathcal{A}_\infty = \bigcup_{\tau,E < \infty} \mathcal{A}_{\tau,E}. \quad (1)$$

Here $\tau, E \geq 0$; the union permits arbitrarily large finite resources, not an experiment completed at infinite time. Costs of a complete adaptive protocol must include every branch used in its certificate. Known gauge redundancies should already be identified in the declared carrier. The observer, model class, allowed preparations and cost convention are part of the access record, not inferred from a response quotient. If feasibility depends on W , taking the intersection of feasible labels over C defines a conservative common class. Its horizon need not survive model-dependent feasible actions, learning, or an enlarged physical carrier. An informative failure must be represented as an outcome, not silently deleted.

$$g_{\tau,E}(W, V) = \sup_{p \in \mathcal{A}_{\tau,E}} |r_p(W) - r_p(V)|, \quad g_\infty(W, V) = \sup_{p \in \mathcal{A}_\infty} |r_p(W) - r_p(V)|. \quad (2)$$

Set the supremum of an empty protocol family to zero; infinite values are allowed. A pair with $W \neq V$ and $g_\infty = 0$ has an exact horizon relative to this access record. This asserts response equality, not ontological identity. “Unknown” describes missing information; “currently inaccessible” refers to a specified finite budget; “permanently inaccessible” quantifies over the declared union. A missing certificate is not a proof of a horizon.

3 Separation, noise and adaptive protocols

Proposition 3.1 (Finite witnesses and a strict error threshold). *For the access record above,*

$$g_\infty = \sup_{\tau,E < \infty} g_{\tau,E}. \quad (3)$$

A pair is exactly separated by some finite protocol if and only if $g_\infty > 0$. Suppose the complete scalar observation has a deterministic additive error in $[-\varepsilon, \varepsilon]$, with the same $\varepsilon \geq 0$ for each protocol. A single-protocol worst-case certificate exists if and only if $g_\infty > 2\varepsilon$.

Proof. Every protocol belongs to some finite budget, proving both inequalities in (3). A positive supremum contains a positive witness. For a fixed protocol, the two possible observation intervals are disjoint exactly when $|r_p(W) - r_p(V)| > 2\varepsilon$. The definition of supremum supplies a witness above any strictly smaller threshold. At equality the intervals can touch, and if $g_\infty = 2\varepsilon$ no protocol exceeds the threshold. $\square$

This is a worst-case bounded-error statement, not a statistical power theorem. Independent repeated noise, correlations, confidence levels and adaptive stopping need their own model. Tolerance indistinguishability is not generally transitive, as already noted in WR-I: responses $0, \varepsilon, 2\varepsilon$ with tolerance $\varepsilon > 0$ give a counterexample. The compact finite-separation argument belongs to WR-I and is not repeated here.

Proposition 3.2 (Finite adaptive closure). *Fix a finite-depth protocol tree with finite outcome alphabets. Actions and stopping decisions are chosen by one policy whose randomization is independent of the candidate world. Feasible actions and resource certificates are common at each history. If for W, V every common feasible action p at every possible history h has the same conditional outcome law $q_W(y | h, p) = q_V(y | h, p)$, then the full stopped transcript laws agree.*

Proof. The empty-history probability agrees. Inductively a child-history probability is the common parent probability times the common policy action probability times the common conditional response probability. Stopping adds a common factor and can be represented by an absorbing label. Induction through the finite depth proves equality of every transcript probability. $\square$

Equality of isolated one-step marginals is insufficient: two bits can each be fair in both worlds but always equal in one and always opposite in the other. A two-step transcript distinguishes them. Likewise, two individually affordable actions need not be jointly affordable. The proposition requires conditional laws and costs of complete policies; it does not justify enlarging a static family without an audit.

Example 3.3 (Pairwise access is not uniform access). On $C = [0, \infty)$ use $r_T(\theta) = (T - \theta)_+$ for finite $T \geq 0$, in dimensionless toy units. Every two distinct parameters have a separating finite T . For every fixed finite deadline T_* , however, all $\theta \geq T_*$ give zero responses for all $T \leq T_*$. Thus no finite deadline identifies the whole noncompact carrier. This does not contradict WR-I's compact-domain result.

4 A causal winding probe

Use a prescribed flat FLRW metric with $a \in C^1([0, \infty))$, $a > 0$, $a(0) = 1$:

$$ds^2 = -c^2 dt^2 + a(t)^2 d\mathbf{x}^2, \quad S(T) = c \int_0^T \frac{dt}{a(t)}, \quad S_\infty = \lim_{T \rightarrow \infty} S(T). \quad (4)$$

Coordinates $\mathbf{x}$ and S have length units; a is dimensionless and c has units length/time. Compare spatial $\mathbb{R}^3$ with $\mathbb{T}_L^3 = \mathbb{R}^3 / (L\mathbb{Z})^3$, $L > 0$, under the same a . A comoving laboratory at the origin launches an ideal tagged null ray at $t = 0$ along a known lattice axis. Only its own first return is read. There are no mirrors, scattering, absorption or false returns. In $\mathbb{R}^3$ this ray never returns. Axis knowledge, tag recognition and collection are supplied calibration assumptions, not inferred topology.

The remaining conformal span is the standard comoving event-horizon distance when finite, for this observer and future history; it is not the Hubble radius or the past particle horizon [10]. Topological multiple images and covering-space constructions are established cosmological tools [11]. We use an explicit ideal active protocol rather than an actual catalogue or CMB inference.

Theorem 4.1 (Strict first-return criterion). *The torus ray has a unique finite first-return time T_L if and only if $L < S_\infty$. When it exists, $S(T_L) = L$. If $L \geq S_\infty$, it gives the same no-return response as the Euclidean ray at every finite deadline.*

Proof. The null equation gives $dx/dt = c/a(t)$ on the lifted axis, hence $x(T) = S(T)$. The first positive lattice point is L . Since $S'(T) = c/a(T) > 0$, S is continuous and strictly increasing from zero. It reaches every $0 < L < S_\infty$ exactly once. For every finite T , the positive tail integral implies $S(T) < S_\infty$ when the limit is finite; if it is infinite the conclusion is immediate. Equality $L = S_\infty$ therefore gives no finite return. $\square$

The length is L , not the injectivity radius $L/2$ and not the $2R$ patch condition of WR-VIII. Arbitrary directions need not close; an unknown orientation requires additional protocols. No-return equivalence here is relative to the declared tagged winding observations. It does not prove agreement of all global classical fields, mode spectra, quantum states or imaginable experiments. In particular, no-return at one deadline does not distinguish a large torus from its cover.

5 A time–energy detection frontier

Now impose a nondecreasing, a launch cap $E_{\max} \geq 0$, and a detector threshold $E_d > 0$, both in joules. The declared geometric-optics packet model conserves photon number and gives collected energy $E_{\text{arr}} = E_0/a(T_L)$ for a launch energy E_0 at $a(0) = 1$, using the standard cosmological redshift relation [10]. Collection of the entire packet, no losses and a deterministic inclusive detector threshold are idealizations. The cap excludes preparation and laboratory overhead; it is not a universal total-energy bound. No diffraction, backreaction, quantum detection model or realizable apparatus is certified.

Proposition 5.1 (Conditional resource certificate). *There exists a detectable first return by deadline $\tau < \infty$ with $0 \leq E_0 \leq E_{\max}$ if and only if*

$$L < S_\infty, \quad T_L \leq \tau, \quad E_{\max} \geq E_{\text{req}}(L) := E_d a(T_L). \quad (5)$$

Proof. Theorem 4.1 supplies the unique return. Detection requires

$$E_0/a(T_L) \geq E_d \iff E_0 \geq E_d a(T_L).$$

Such an energy lies under the cap exactly under (5); choose $E_0 = E_{\text{req}}$. Both finite-deadline and detector equalities are included by their stated conventions. $\square$

Corollary 5.2 (Exponential benchmark). *For $a(t) = e^{Ht}$, $H > 0$ in inverse seconds, set $x = HL/c$. For $0 < x < 1$,*

$$S(T) = \frac{c}{H}(1 - e^{-HT}), \quad S_\infty = \frac{c}{H}, \quad T_L = -\frac{1}{H} \log(1 - x), \quad E_{\text{req}} = \frac{E_d}{1 - x}. \quad (6)$$

For $E_{\max} \geq E_d$, the detectable positive periods at finite budget are exactly

$$0 < L \leq L_{\text{det}}(\tau, E_{\max}) = \frac{c}{H} \min\{1 - e^{-H\tau}, 1 - E_d/E_{\max}\}. \quad (7)$$

The set is empty for $E_{\max} < E_d$, and also when the right endpoint is zero.

Proof. Integrate (4) and solve $S(T_L) = L$. Then $a(T_L) = (1 - x)^{-1}$. Substituting into (5) gives $x \leq 1 - e^{-H\tau}$ and $x \leq 1 - E_d/E_{\max}$. Nondecreasing $a \geq 1$ excludes detection when $E_{\max} < E_d$. $\square$

For $H\tau = \log 2$ and $E_{\max} = 2E_d$, the endpoint $L = c/(2H)$ is detected exactly at both equalities. At $L = c/H$ the return never occurs at finite time, even with unlimited launch energy. As $x \uparrow 1$, T_L diverges logarithmically and E_{req} diverges as $(1 - x)^{-1}$. A fixed energy cap creates a stricter detection limit than causal reach. These are different reasons for identical no-detection responses.

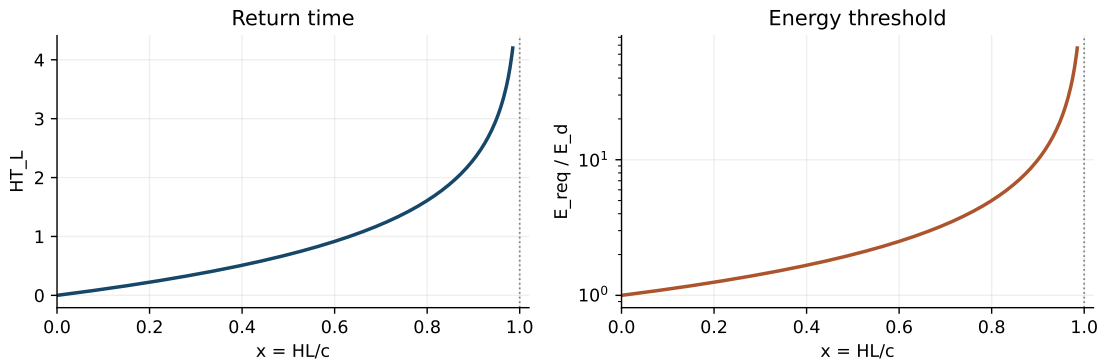


Figure 1: Synthetic exponential model: normalized return time and required energy. The causal endpoint $x = 1$ is excluded; finite resource thresholds may be attained. No cosmological data are fitted.

6 Current compatibility does not fix future reach

Theorem 6.1 (Smooth agreement with different tail convergence). *Fix $H > 0$, $T_* > 0$ and $\delta > 0$. There are positive, strictly expanding C^∞ scale factors on $\mathbb{R}$ which both equal e^{Ht} for every $t \leq T_*$, but have finite versus infinite S_∞ from $t = 0$.*

Proof. Let $\rho(u) = e^{-1/u}$ for $u > 0$ and zero otherwise, and set

$$s(u) = \frac{\rho(u)}{\rho(u) + \rho(1-u)}, \quad h_-(t) = \begin{cases} H, & t \leq T_*, \\ H[1 - s((t - T_*)/\delta)] + \frac{Hs((t - T_*)/\delta)}{1 + H(t - T_*)}, & t > T_*. \end{cases} \quad (8)$$

The denominator defining s is positive everywhere; $s = 0$ on $u \leq 0$, $s = 1$ on $u \geq 1$, and it is smooth and flat at both join points: every derivative of $e^{-1/u}$ is a polynomial in $1/u$ times $e^{-1/u}$ and tends to zero as $u \downarrow 0$. Hence h_- is smooth and positive, agrees with H through T_* , and equals $H/[1 + H(t - T_*)]$ after $T_* + \delta$. Define

$$a_+(t) = e^{Ht}, \quad a_-(t) = \exp\left(\int_0^t h_-(v) dv\right). \quad (9)$$

Both have the stated past and $\dot{a}_\pm/a_\pm > 0$. For $t \geq T_* + \delta$,

$$a_-(t) = A[1 + H(t - T_*)], \quad A = \frac{a_-(T_* + \delta)}{1 + H\delta} > 0. \quad (10)$$

Its reciprocal has a divergent logarithmic tail integral, whereas a_+ gives $S_\infty = c/H$. Smoothness also gives agreement of all derivatives at T_* . $\square$

In the calibrated flat current carrier, both histories have $H(z) = H$ and $D(z) = cz/H$ on any supplied finite $[0, Z]$. A period $L \geq c/H$ has no finite winding return under a_+ but does return under a_- , at some finite time and finite threshold energy. To compare topology lifts on the restricted current patch of WR-VIII, also impose $L > 2cZ/H$; choosing $L > \max\{c/H, 2cZ/H\}$ meets both requirements. Equality of other current fields still requires the explicit patch-copy premise. Until T_* the backgrounds agree exactly. This is a change of future tail, not a compact bump. A prescribed matter evolution law, analyticity, or a fixed cosmological parameter family can exclude this construction; none is supplied here. It is a kinematic counterexample to inferring permanent access from current background responses, not a pair of solutions of one declared matter theory.

Proposition 6.2 (Compact future changes preserve convergence type). *If positive continuous a, b agree for every $t \geq T_1 < \infty$, then $\int_0^\infty dt/a(t)$ is finite if and only if $\int_0^\infty dt/b(t)$ is finite.*

Proof. Each reciprocal is bounded on the compact interval $[0, T_1]$ because its denominator has a positive minimum. The remaining tails are identical. $\square$

The numerical value of a finite span can still change; a fixed L can cross that value. The compact future bumps of WR-VIII therefore establish temporal ambiguity but cannot alone switch finite to infinite conformal reach. Theorem 6.1 supplies the missing noncompact-tail construction.

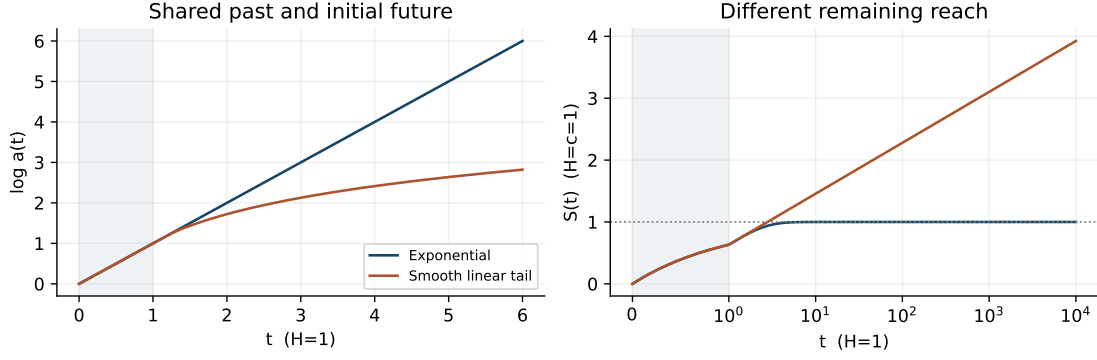


Figure 2: Synthetic $H = c = 1$, $T_* = 1$, $\delta = 1/2$. The histories and reach agree until the switch; thereafter the exponential span approaches one while the smooth linear-tail span is unbounded. Finite plot values do not prove divergence; Eq. (10) does.

7 Unknown futures and the order of quantifiers

Proposition 7.1 (Possible versus excluded winding access). *Let $\mathcal{G} \neq \emptyset$ be a declared family of positive future scale factors as in Section 4. A fixed period $L > 0$ has no finite causal winding return for any $\gamma \in \mathcal{G}$ if and only if*

$$L \geq \sup_{\gamma \in \mathcal{G}} S_{\infty}^{(\gamma)}. \quad (11)$$

A return is possible in at least one future exactly when some $S_{\infty}^{(\gamma)} > L$. Return in every future requires $L < S_{\infty}^{(\gamma)}$ for every γ ; it does not imply a common finite resource certificate.

Proof. Apply Theorem 4.1 to each future. Having $L \geq S_{\infty}^{(\gamma)}$ for all γ is exactly (11), with the usual extended supremum. The last claim follows from the example below. $\square$

Example 7.2 (All returns exist, but no uniform budget). Take $H_n = H_0(1 - 1/n)$, $a_n = e^{H_n t}$, $n \geq 2$, and $L = c/H_0$. Every history has $S_{\infty}^{(n)} = L/(1 - 1/n) > L$, while their infimum equals L and is not attained. Equations (6) give

$$T_n = \frac{\log n}{H_0(1 - 1/n)} \longrightarrow \infty, \quad E_{\text{req},n} = nE_d \longrightarrow \infty. \quad (12)$$

Thus $\forall \gamma \exists (\tau, E)$ cannot be replaced by $\exists (\tau, E) \forall \gamma$. Requiring $L < \inf_{\gamma} S_{\infty}^{(\gamma)}$ would incorrectly exclude this example. These H_n are a family with uncertain current rate, not the exactly shared current data of Theorem 6.1.

8 Scientific ceiling and closure of the cycle

The central distinction is between a response class at current resources and a horizon relative to a justified future access record. In the benchmark, the record explicitly fixes future expansion, lattice orientation, packet transport, detector threshold and allowed launches. Changing any of these can change the certificate. A causal horizon restricts information propagation; it does not say that no mathematical or physical structure exists beyond it. Exact equality of the winding response says nothing about unrelated protocols.

A mathematical FLRW history is not automatically a solution of a prescribed matter theory or a realistic universe. A usable physical analysis must add a well-posed matter/gravity model, perturbations and fields, initial and boundary data, observer survival, apparatus dynamics, directional uncertainty, losses and statistical detection. The current article certifies none of these. Known event-horizon and redshift formulas, torus coverings, bounded-error intervals and conditional-product arguments are prior tools. The

contribution is their explicit WREH access/resource contract and the future-tail counterexample; broader priority remains unestablished.

This is the ninth and final technical article of the manifesto’s first cycle. It does not declare all nine articles published, reviewed or merged. WR-V remains a separate review draft; WR-VII publication is author-reported pending deposited-file verification; WR-VIII v0.2 is deposited at DOI 10.5281/zenodo.23266980. Earlier frozen sources are retained. Open PR dependencies and scientific review remain separate from the completion of this draft. A later synthesis requires its own scope and evidence. WREH and Boundary Compensation remain distinct programmes.

Reproduction and audit. The package contains shared bilingual LaTeX, analytic figures, an offline demonstration, symbolic/numerical and adversarial boundary checks, a source map and an audit. Computation checks finite witnesses and implementations; the proofs establish the infinite-tail claims. AI-assisted drafting and automated technical audit are disclosed. They are not independent external scientific peer review, and no journal acceptance or WR-IX deposit is claimed.

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